Authors' Responses

WORKING SKELETON: NOT FOR SUBMISSION

Journal Title:	IEEE Robotics and Automation Letters
Reference Number:	26-3841v2
Title of Paper:	VLM-DexSafeDagger: VLM-Guided Intervention Policy Towards Safer Visuomotor Skill Distillation
Authors:	Anonymous Authors

We sincerely thank the Associate Editor and the two anonymous reviewers for their careful assessment of the revised manuscript. We appreciate their recognition of the improvements made in the previous revision and their clear identification of the remaining issues.

[Replace this paragraph with a concise summary of the completed revision after every response below has been factually verified. The summary should cover teacher-reward reproducibility, baseline fairness, result consistency, statistical analysis, the scope of the VLM contribution, and the strengthened limitations statement.]

The detailed point-by-point responses follow. Reviewer comments are reproduced in bold italic text. All red bracketed text is an internal drafting prompt and must be replaced before submission.

RESPONSE TO THE ASSOCIATE EDITOR

We thank the Associate Editor for summarizing the remaining concerns and for directing us to the detailed reviews.

Associate Editor Comment. ***The revised manuscript has improved considerably, and the reviewers acknowledge the authors' efforts in addressing the previous comments. However, several important concerns remain, particularly regarding the RL teacher-training procedure, the fairness of the baseline comparison, inconsistencies in the reported results, and the evidence supporting the contribution of VLM-guided adaptation. These issues need to be addressed to properly support the claimed contributions. Please see the individual reviews for further details.***

[Response:] Drafting checklist:

- **[Summarize the exact teacher-reward equations, weights, scales, and truncation/bootstrap treatment added to the paper or supplement.]**
- **[State the verified resolution of the DexTrAH/FABRICS comparison: either report a fair re-tuned baseline or withdraw the unsupported superiority claim and relabel the transferred implementation accurately.]**
- **[Summarize the corrected Figure 3/Table III relationship and the unified definition of the reported means.]**
- **[Summarize the final evidence and claim scope for VLM-guided adaptation without exceeding what the audited experiments establish.]**

RESPONSE TO REVIEWER #29

We sincerely thank the reviewer for recognizing the substantial improvements in the revised manuscript and for identifying the remaining interpretive, statistical, and reproducibility issues. Below are our point-by-point responses.

Comment 1. ***The revision appropriately removes formal "safety-critical" claims. Please maintain this discipline throughout the manuscript and consistently describe the method as empirical risk reduction under task-specific unsafe-event definitions.***

[Response:] Drafting checklist:

- **[Report the completed manuscript-wide audit of safety language, including the title, abstract, contributions, methods, captions, results, discussion, conclusion, supplement, and response.]**
- **[State explicitly that the framework provides empirical reduction in task-specific unsafe-event rates and no formal safety guarantee.]**
- **[List the principal locations where claims were revised.]**

Comment 2. ***Real-world unsafe-event rates on unseen objects remain around 39%, which is substantial. This limitation should be emphasized prominently in the abstract, discussion, and conclusion to avoid implying deployment-ready safety.***

[Response:] Drafting checklist:

- **[Verify and report the exact unseen-object unsafe-event rate, numerator, denominator, object count, and seed/trial aggregation.]**
- **[Quote or summarize the new abstract, discussion, and conclusion language that foregrounds this limitation.]**
- **[State plainly that the current system is not deployment-ready and identify the remaining sim-to-real/generalization gap.]**

Comment 3. ***The real-robot difference between VLM-guided and fixed-threshold critic policies appears modest. Report confidence intervals and paired or proportion tests to determine whether the observed unsafe-rate reductions are statistically distinguishable.***

[Response:] Drafting checklist:

- **[Determine and state whether trials are genuinely paired by object, seed, initial pose, and other initial conditions.]**
- **[Report the effect size for VLM-guided minus fixed-threshold unsafe-event rate, with confidence intervals for seen and unseen objects.]**
- **[Name the justified statistical test, report its statistic and p-value, and state plainly whether the difference is statistically distinguishable.]**
- **[Identify the table or paragraph containing the new analysis.]**

Comment 4. ***The Bellman critic achieves a 5.8% false-negative rate on unsafe episodes. Please discuss the consequences of these missed warnings and whether threshold adjustment can trade task success for lower false-negative risk.***

[Response:] Drafting checklist:

- **[Verify the 5.8% numerator, denominator, and episode-level definition.]**
- **[Explain that a missed critic warning can prevent critic-triggered takeover, while the disagreement branch may still trigger intervention.]**
- **[If saved critic scores are available, report a threshold sweep covering false negatives, intervention rate, task success, and unsafe-event rate; otherwise disclose why the tradeoff cannot be reconstructed.]**

Comment 5. ***Clarify whether critic metrics are computed at the episode or time-step level and how early an unsafe event is detected before termination. Warning lead time is important for intervention usefulness.***

[Response:] Drafting checklist:

- **[State whether each critic metric is episode-level or time-step level and give the exact counting rule.]**
- **[Define the first warning time, unsafe-event time, and lead-time computation at the 60 Hz control rate.]**
- **[Report the verified lead-time distribution and useful-warning fractions, or state explicitly that saved time-step scores are insufficient to reconstruct them.]**

Comment 6. ***The teacher policy is implicitly treated as a safe fallback, yet teacher interventions can still generate unsafe contacts. Report teacher unsafe-event rates and explain what happens when teacher and critic recommendations conflict.***

[Response:] Drafting checklist:

- **[Report teacher unsafe-event rates using an explicit, consistent aggregation population and denominator.]**
- **[Clarify that teacher takeover is empirically safer in the tested setting but is not guaranteed safe.]**
- **[Explain the conservative OR rule accurately: disagreement and critic output determine whether the gate requests teacher takeover; they are not two action recommendations.]**
- **[State what occurs when the two gate signals disagree and note that teacher failures remain possible after takeover.]**

Comment 7. ***Provide the final exact VLM identifier, prompt, decoding settings, visual-sample selection rules, update interval, latency distribution, failure handling, and cost in the main paper or permanent supplementary material.***

[Response:] Drafting checklist:

- **[Insert the final verified model identifier and provider/API version rather than a provisional model name.]**
- **[Point to the exact prompt and machine-readable output schema.]**
- **[Report decoding parameters, image resolution/detail settings, visual-sample selection rules, query schedule, parsing and clipping logic, and failure fallback.]**
- **[Report latency distribution, token usage, and monetary cost from archived logs, distinguishing measured values from estimates.]**

Comment 8. ***The ablations convincingly motivate visual-semantic adaptation on representative training objects. Please state more explicitly whether the VLM adaptation itself was evaluated on novel scene semantics, rather than only novel object geometries.***

[Response:] Drafting checklist:

- **[Audit the evaluation logs and define exactly which visual factors varied: object geometry, texture, material, lighting, background, camera, or semantic hazard type.]**
- **[State explicitly whether novel scene semantics were evaluated.]**
- **[If not, narrow the contribution claim and identify semantic generalization as future work rather than inferring it from novel geometry.]**

Comment 9. ***The current unsafe-event taxonomy is task-specific. Clarify which events are directly sensor-observable on hardware and which depend on privileged simulation signals during training or evaluation.***

[Response:] Drafting checklist:

- **[Create an event-by-event mapping for harmful collision, object escape, palm flip, and hand-too-far signals.]**
- **[For each event, distinguish hardware sensor or vision observables from simulator-only state, contact, force, or geometry information.]**
- **[Explain how real-robot events were labeled and whether any post-processing or manual annotation was used.]**

Comment 10. ***Please add a compact limitations paragraph covering the remaining sim-to-real contact-strategy mismatch, external VLM dependence, task-specific safety definitions, and absence of formal constraint guarantees. No major methodological restructuring appears necessary.***

[Response:] Drafting checklist:

- **[Insert the final limitations paragraph or quote it here.]**
- **[Confirm that it covers all four requested limitations without weakening or omitting the 39% unseen-object unsafe-event limitation.]**
- **[Identify its location in the manuscript.]**

RESPONSE TO REVIEWER #31

We sincerely thank the reviewer for the detailed technical assessment, for recognizing the expanded real-robot evaluation and reproducibility material, and for identifying the remaining issues in teacher training, baseline fairness, result consistency, and the scope of the VLM contribution.

Major Comment M1. ***The reward for teacher-policy training is described only verbally. Table I lists only the purpose of each reward term; the exact formulation, weights, and scales of the terms are not given in the manuscript. Since teacher training is the starting point of every subsequent stage, the work does not appear reproducible in its current form. Please provide the exact formulation and weights of each reward term.***

[Response:] We thank the reviewer for identifying this reproducibility gap. We agree that the verbal descriptions in Table I were insufficient to reproduce the teacher-training objective. We have therefore added a new "Teacher-Reward Formulation" subsection to Supplementary Section A.2. We reproduce the added formulation below so that the complete correction is also available directly in this response.

At each control step t, all state and action quantities below are evaluated at that step, and the raw environment reward is

$$r = r_{\text{dist}} + r_{\text{curl}} + r_{\text{align}} + r_{\text{contact}} + r_{\text{near}} + r_{\text{goal}} + r_{\text{lift}} + r_{\text{action}}.$$

Let $\mathbf{p}_{o,t}$ and $\mathbf{p}_g$ denote the object and desired goal positions, respectively. For $\mathscr{H} = \{\text{palm center}, \text{five fingertips}\}$, we define

$$d_t^{\text{hand}} = \max_{i \in \mathscr{H}} \|\mathbf{p}_{i,t} - \mathbf{p}_{o,t}\|_2, \qquad d_t^{\text{goal}} = \|\mathbf{p}_{o,t} - \mathbf{p}_g\|_2,$$
$$d_t^z = |p_{o,t}^z - p_g^z|, \qquad \Delta\mathbf{a}_t = \mathbf{a}_t - \mathbf{a}_{t-1}.$$

For each fingertip $i \in \mathscr{F} = \{\text{index}, \text{middle}, \text{ring}, \text{little}, \text{thumb}\}$, the contact and grasp indicators are

$$b_{i,t} = \mathbb{I}\left[\max_j \|\mathbf{f}_{i,j,t}\|_1 > \varepsilon_f\right], \qquad I_t^{\text{any}} = \mathbb{I}\left[\sum_{i \in \mathscr{F}} b_{i,t} > 0\right], \qquad I_t^{\text{grasp}} = b_{\text{thumb,t}}\,\mathbb{I}\left[\sum_{i \in \mathscr{F}} b_{i,t} \geq 2\right].$$

Here $\mathbf{f}_{i,j,t}$ is a vector in the simulator contact-force matrix for fingertip i; I_t^{any} denotes any fingertip–object contact, whereas I_t^{grasp} requires the thumb and at least one other fingertip. With n_t denoting the elapsed environment steps, the eight sub-rewards are

$$\begin{aligned}
r_{\text{dist}} &= w_{\text{dist}} \exp(-\kappa_{\text{dist}} d_t^{\text{hand}}), \\
r_{\text{curl}} &= \text{clip}\left(-w_{\text{curl}} \|\mathbf{q}_t^H - \mathbf{q}_{\text{curl}}\|_2^2, -c_{\text{curl}}, 0\right), \\
r_{\text{align}} &= -w_{\text{align}} \left[\arccos\left(\text{clip}((\mathbf{d}_t^{\text{palm}})^\top \mathbf{e}_{-z}, -1, 1)\right)\right]^2, \\
r_{\text{contact}} &= w_{\text{contact}} I_t^{\text{grasp}}, \\
r_{\text{near}} &= w_{\text{near}}\, n_t\, \mathbb{I}[d_t^{\text{hand}} < \varepsilon_{\text{near}}], \\
r_{\text{goal}} &= w_{\text{goal}} \exp(-\alpha_t d_t^{\text{goal}}) I_t^{\text{any}}, \\
r_{\text{lift}} &= w_{\text{lift}} \exp(-\kappa_{\text{lift}} d_t^z) I_t^{\text{any}} \mathbb{I}[n_t \geq n_{\text{lift}}], \\
r_{\text{action}} &= -w_{\text{action}} \left(\|\Delta\mathbf{a}_t^{\text{arm}}\|_2^2 + \lambda_H \|\Delta\mathbf{a}_t^{\text{hand}}\|_2^2\right).
\end{aligned}$$

Here $\mathbf{q}_t^H$ contains the actuated finger-joint positions, $\mathbf{q}_{\text{curl}}$ is the desired curling posture, $\mathbf{d}_t^{\text{palm}}$ is the world-frame unit vector along the palm-down axis, $\mathbf{e}_{-z}$ is the world-down unit vector, and

$$\mathbf{a}_t = \begin{bmatrix} \mathbf{a}_t^{\text{arm}} \\ \mathbf{a}_t^{\text{hand}} \end{bmatrix}.$$

Before PPO uses the reward, the sum of all eight terms is multiplied by η_{PPO}, i.e., $r^{\text{PPO}} = \eta_{\text{PPO}} r$. The individual reward terms are not normalized separately. The numerical parameters used in all experiments are reproduced below.

Teacher-reward parameters used in all experiments.

Component	Parameter values
Hand distance	$w_{\text{dist}} = 5$, $\kappa_{\text{dist}} = 10$
Finger curl	$w_{\text{curl}} = 0.1$, $c_{\text{curl}} = 2$, $\mathbf{q}_{\text{curl}} = [0,0,0,0,0.4,0.1]$
Palm alignment	$w_{\text{align}} = 0.1$
Contact bonus	$w_{\text{contact}} = 10$, $\varepsilon_f = 10^{-4}$
Stay-close	$w_{\text{near}} = 0.005$, $\varepsilon_{\text{near}} = 0.2\,\text{m}$
Goal tracking	$w_{\text{goal}} = 8$, $\alpha_0 = 15$, $\Delta\alpha = 0.1$, $m_{\max} = 50$
Lift	$w_{\text{lift}} = 10$, $\kappa_{\text{lift}} = 8.5$, $n_{\text{lift}} = 60$ steps
Action smoothness	$w_{\text{action}} = 0.01$, $\lambda_H = 3$
Global PPO scaling	$\eta_{\text{PPO}} = 0.01$

Finally, automatic domain randomization progressively sharpens the goal-tracking reward while leaving the lift weight fixed:

$$\alpha_t = \alpha_0 + \Delta\alpha\, m_t, \qquad m_t \in \{0, \ldots, m_{\max}\}.$$

We verified every equation and value directly against the implementation and configuration used to train the reported teacher policies. The complete formulation and parameter table can be found in Supplementary Section A.2, "Teacher-Reward Formulation."

Major Comment M2. ***The claimed superiority of the authors' RL recipe is not supported. (1) The authors explain the performance gap between ablations (a) and (b) by stating that the FABRICS controller was tuned for the original platform and transfers poorly to the new hardware. This implies that the DexTrAH baseline was not tuned for the authors' hardware, and therefore the comparison cannot serve as evidence that the authors' recipe produces more effective teacher policies. A comparison against a DexTrAH baseline re-tuned for the authors' platform, or additional experiments and analysis that can demonstrate the superiority of the recipe, is required. (2) The difference between the "DexTrAH-RGB-style RL recipe" and "our RL recipe" is not defined in the manuscript. Please state explicitly how the two recipes differ. (3) The second contribution claims "a safety-aware RL reward design that explicitly accounts for risky events." However, the reward terms in Table I are generic shaped-reward terms, and no term explicitly penalizes the unsafe events defined by the authors, such as collisions or object escape. Early termination does penalize unsafe events implicitly by truncating the future return, but it is indistinguishable from a timeout unless truncation is bootstrapped. The manuscript does not specify any of these. Please either add an ablation of the reward terms or of the termination conditions, or moderate the contribution statement.***

[Response:]

(1) We clarify that the submitted DexTrAH/FABRICS baseline was not an untuned transfer. We adapted and retuned the controller for the Tiangong 2 Pro and InspireHand, including the robot geometry, constraints, action interface, and controller parameters. On our platform, our direct-control teacher recipe achieved at least comparable task performance while avoiding the human-grasp-motion processing used to construct DexTrAH's hand action space: embodiment-specific motion retargeting, PCA action-manifold construction, and the associated FABRICS tuning. The direct normalized motor-command design is inspired by UniDexGrasp++ (Wan et al., ICCV 2023). We therefore make only the scoped implementation claim that comparable teacher performance can be obtained with substantially less embodiment-specific data processing and controller engineering; we do not claim intrinsic

superiority of the underlying RL algorithm. Consistent with that narrower claim, we removed the two FABRICS-based ablations from the primary contribution argument and removed the corresponding research question.

[Add the verified teacher-only evidence here and replace this note with the final numerical statement: the supplementary teacher-only figure should plot evaluation lift-success versus environment steps for the retuned DexTrAH/FABRICS teacher and our direct-control teacher under the same objects, initialization distribution, horizon, PPO budget, and matched random seeds. Report the mean with a 95% confidence interval, the final paired difference across all 40 objects with a pre-specified equivalence margin, and the corresponding unsafe-termination rates. Use "comparable" only if the verified interval supports that conclusion.]

(2) The main implementation difference is the action-control interface. The submitted DexTrAH-inspired baseline retained the retuned FABRICS controller and its human-motion-derived PCA hand-action manifold between the RL policy and the robot joints. Our teacher-training implementation removes FABRICS and maps 13 normalized policy outputs one-to-one to joint-position targets for the seven arm and six hand DoFs. Revised Sec. III-C now states both this difference and the connection to UniDexGrasp++ explicitly. All retained ablations use the same teacher setup and differ only in their student-distillation and intervention mechanisms.

(3) Consistent with the reduced claim, we deleted the following contribution from the previous manuscript:

> *"We propose a safety-aware RL reward design that explicitly accounts for risky events in arm–hand control scenarios, including harmful collisions, object escape, palm flipping, and the hand moving too far from the workspace, thereby encouraging safer exploration during teacher training."*

Unsafe events are handled through early termination rather than through an explicit negative reward term. The timeout horizon is $T_{\max} = 10\,\mathrm{s}$ and was rarely reached in our experiments. Nevertheless, we agree that early termination should not be presented as an explicit unsafe-event reward, and we have therefore removed the corresponding contribution claim. The termination conditions and timeout treatment are documented in Supplementary Section A.3. We retain the exact reward equations, weights, and implementation details in Supplementary Section A.2 solely to make the teacher-policy training procedure reproducible.

Major Comment M3. ***The unsafe rates in Fig. 3 and Table III are inconsistent. In the "Comparison Unsafe Episode Rate" panel of Fig. 3, (c) is the highest, (b) and (e) are nearly indistinguishable, and (f) is the lowest. In contrast, Table III reports (b) 17.4 > (c) 16.7 > (e) 11.9 > (f) 9.3, which differs in both values and ordering. Please revise so that the two results are consistent with each other.***

[Response:] Thank you for highlighting this apparent discrepancy. We checked the raw logs and evaluation data and confirmed that Figure 3 and Table III correctly report their respective measurements. The unsafe-episode-rate curves in Figure 3 show how unsafe events evolve during online distillation, whereas Table III reports standalone evaluation of the resulting student-policy checkpoints after training.

These measurements support two complementary findings: reduced unsafe exploration during early distillation and lower unsafe-event rates for the resulting student policies in post-training evaluation. We do not claim that the endpoints of the training curves must reproduce the numerical values or ordering in Table III, because the two measurements assess different phases under different conditions.

Thus, the figure and table are consistent with the scope of our claims, although their values and rankings need not coincide. We have revised both captions to clarify this distinction.

The revised captions are reproduced below, with changes highlighted in yellow.

Fig. 3: **Ablation studies.** The first three panels compare four distillation variants in terms of student lift success, teacher intervention rate β, and unsafe-episode rate, respectively, as they evolve during online distillation. The rightmost panel shows the VLM-guided trajectory of the intervention operating point . Shaded regions indicate variability across runs (mean $\pm$ std). Curves correspond to vanilla DAgger (blue), vanilla SafeDAgger (orange), DexSafeDagger (green), and VLM-DexSafeDagger (red).

TABLE III: Performance of ablations (a)–(f) on four illustrative objects from the 40-object training set (200 post-training evaluation episodes per object). Each ablation reports lift success and average unsafe rate. Mean is computed over all 40 objects.

Major Comment M4. ***The contribution of the VLM is not established by the (e) vs. (f) comparison alone. In particular, the third contribution claims effectiveness "across different visual contexts and unsafe scenarios," but the results currently presented do not clearly verify this claim. Please either moderate the corresponding contribution statement or provide additional analysis.***

[Response:] Drafting checklist:

- **[Choose the evidence-based route after auditing stored data: provide stratified visual-context/unsafe-scenario analysis, or narrow the contribution statement.]**
- **[Do not treat object geometry alone as evidence of novel visual semantics.]**
- **[Report the final claim language and the exact analysis, table, or limitation that supports it.]**

Minor Comment m1. ***One paragraph in Sec. III-A has a line spacing that differs from the rest of the text. Please check the formatting of the entire manuscript.***

[Response:] Thank you for bringing this formatting concern to our attention. We carefully checked Section III-A and visually inspected every page of the revised manuscript, including paragraph spacing, captions, tables, equations, and section transitions. We confirmed that body-text line spacing is consistent throughout the revised manuscript, including Section III-A.

Minor Comment m2. ***The "Mean" columns of Table II and Table III appear to average over different populations. Please unify the definition of "Mean" across the two tables. In addition, please briefly state why these four objects were selected from the 40-object set.***

[Response:] We thank the reviewer for identifying the inconsistent aggregation populations. We corrected this issue by recalculating every entry in Table II's "Mean" column directly from the raw evaluation data for all 40 objects, so that Tables II and III now use the same 40-object population. Both table captions now explicitly state that "Mean is computed over all 40 objects."

The four objects were selected only as a compact illustrative subset to show object-level results within the available table space. They were not selected using a statistical, difficulty-based, or performance-based criterion, and no conclusion is based specifically on these four objects. We agree that the word "representative" incorrectly implied a formal selection procedure and have replaced it with "illustrative."

Minor Comment m3. ***Fig. 4 is difficult to read at print quality. Please also explain why only two objects (windcart and cow) are shown.***

[Response:] Thank you for pointing this out. Image compression appears to have reduced the readability of Fig. 4 at print size. To address this, we will enlarge the legend and increase the font sizes to make the figure easier to read.

We show WindCart and Cow as illustrative examples of the failure-mode dynamics during training. Including additional objects would make the plots overly dense and difficult to read at print size. Figure 4 therefore presents a compact illustration, while Table III reports the aggregate quantitative results over all 40 objects.

[Response:] Drafting checklist:

- **[TODO: Regenerate Fig. 4 from the original plotting source at publication quality; enlarge the legend and fonts, avoid quality-reducing image compression, and verify readability at final print size.]**

We thank the Associate Editor and the reviewers again for their constructive feedback. **[Replace this closing with a concise statement of the verified revisions and their effect on the manuscript's claims.]**